AI-Based Intelligent HR Agent for Automated Resume Screening, Candidate Ranking and Recruitment Optimization using Machine Learning

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Abstract

The sudden surge in the number of resumes submitted by applicants results in a more complicated process of recruitment and screening that, in its turn, requires intelligent automated approaches capable of handling the growing volumes of information. The need arises for companies to adopt intelligent approaches in candidate evaluation and recruitment process automation. In our research, we propose an intelligent AI-based agent named HR Agent that performs intelligent resume screening using modern machine learning algorithms, ranks resumes according to some predefined criteria and automates the recruitment process completely.

Various machine learning algorithms are used to perform candidate resume classification including such techniques as Random Forest, Logistic Regression, Support Vector Machine, K-Nearest Neighbours, Gradient Boosting and Multi-layer Perceptron. Apart from this, there is a special preprocessing pipeline designed to increase the performance of models. This pipeline uses feature engineering and scaling, outlier detection and winsorization, SMOTE-based class balancing, and controlled noise injection.

Apart from the classification step, the agent uses the heuristic algorithm to rank resumes in accordance with their fit which takes into account the match between the candidate's skills and position requirements, presence of common keywords in the resume and the job description, level of candidate's experience, diversity of skills, and many other criteria. What is more, there is an intelligent recruitment pipeline implemented that includes

Some of these features include automatic scheduling of interviews, and creation of skill-oriented technical and situational interview questions.

Lastly, different measures are employed to evaluate the performance of these models, which include accuracy, precision, recall, F1 score,

area under ROC curve (ROC-AUC), and so forth. It is found that ensemble models like random forest and gradient boosting provide outstanding results.

Keywords

Machine Learning, Artificial Intelligence, Resume Screening, Candidate Ranking, Recruitment Automation, Classification Models, HR Analytics, Optimization Techniques

1. Introduction

Recruitment encompasses all procedures involved in looking for suitable people and selecting those who will take a certain position. With the rapid growth of online job sites, nowadays companies get dozens and sometimes even hundreds of applications for one open position, thus making recruitment a rather complex, time-consuming, and expensive procedure. Conventional recruiting primarily entails manual screening of resumes, which takes a great amount of work and produces inconsistent decisions due to inherent subjectivity.

Modern advances in AI and ML technologies make it possible to create novel recruiting systems that would assist organizations in recruiting more efficiently and effectively. This approach implies automation of the recruitment process and conducting decisions based on relevant information. Different types of recruiting automation software are currently available. However, majority of these solutions are only capable of completing a single task out of many associated with recruiting. Therefore, an overall recruitment system should be created.

Description of the Problem/Gap/Need

This paper suggests the development of automated recruitment process via application of Artificial Intelligence technologies, specifically, development of Human Resources (HR) Agent – an automated system for searching and managing recruiting process. The system analyses resumes and other relevant information provided by candidates, predicts the best candidates for specific positions and ranks them according to their suitability with the description of the jobs. Moreover, scheduling of interviews becomes automated, as well as generation of relevant questions for particular job skills.

Proposed Solution

Multiple machine learning algorithms and models are applied within the HR agent system. They include Random Forest Classifier, Logistic Regression, Support Vector Machine, K-Nearest Neighbours, Gradient Boosting and Multi-Layer Perceptron. In order to make models more efficient and predict candidates correctly, preprocessing pipeline of several different approaches is used. They include scaling and balancing procedures, feature engineering and others. Scoring function for candidate ranking is proposed.

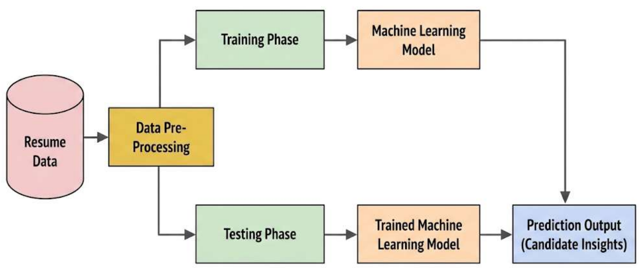

Fig. 1: Basic workflow of AI HR agent

Contributions of this research can be characterized as follows:

• Development of the AI-based automated resume screening and classification system

• Application of multimodal machine learning architecture for better prediction

• Conception and design of heuristic-based ranking systems

• Solution for the whole recruitment pipeline automation

• Optimization and preprocessing techniques application for better model performance

Paper Structure

A literature review will be included in Section II of the paper. Materials and methods will be introduced in Section III. Experimental results will be presented in Section IV. Analysis of experimental results can be found in Section V. Conclusion and future works will be discussed in Section VI.

2. Literature Review

One of the research topics that have been investigated in literature concerning the automation of the recruitment process by means of AI and ML is the application of algorithms and models to automatically analyze potential candidates' qualifications. The approaches that have been proposed in literature for this purpose include, for example, the use of classification models, natural language processing techniques, and recommender systems.

In [1], several learning-based approaches that can be used to solve classification tasks have been considered. Algorithms like logistic regression and support vector machines demonstrated high efficiency in structured data environments. As a consequence, such algorithms became very popular in predicting candidates' qualifications by means of their skills, experience, and education in recruitment systems. Moreover, the random forest technique based on an ensemble of decision trees has been introduced in [2].

It is significant to use NLP in the process of hiring in terms of resume screening and job matching. For instance, in paper [3] intelligent matching system, which uses recommendation technique for matching candidates and job profiles, is presented. Candidate preferences as well as job profile requirements are considered during this procedure. According to previous studies, intelligence matching can increase the efficiency of recruiting. In paper [4], recommendation techniques that are based on machine learning are used in job matching process. It was stated that machine learning recommendation techniques proved to be superior to conventional job matching techniques. Feature engineering and preprocessing techniques are essential components for increasing machine learning model performance. For instance, in paper [5] the significance of data preprocessing techniques, such as normalization, dealing with missing values, and feature selection, was highlighted. As a result, it was revealed that engineering of proper features improves machine learning model performance significantly. Moreover, in paper [6] the significance of optimization techniques, namely regularization and cross-validation, is mentioned in order to prevent overfitting and ensure better performance of machine learning models.

Recently, many researchers try to develop automated systems for the purpose of recruiting. For example, in paper [7], intelligent resume screening system was proposed. This system uses machine learning classification algorithms for sorting out applicants based on their qualification and experience. Automation of hiring tasks will definitely help with recruiting since it allows minimizing human effort. Nevertheless, there is no mention about other techniques than classification.

However, although there have been considerable advancements within this industry, the modern approaches to recruitment appear quite insufficient since all of them are mainly focused on classification and prediction tasks, while none of them has considered ranking, interviewing, and pipeline issues. It should be noted that there has not been enough attention paid to integrating machine learning algorithms with optimization methods.Here, the AI HR Agent appears to be an innovation within this sphere due to its several classification systems, advanced preprocessing technologies, and comprehensive recruiting pipelines.

Table 1: Literature Analysis of Recruitment Methods

Author	Dataset	Approach	Model
Hastie et al.	Stat. data	Supervised ML	LR, DT
Bishop	Pattern data	Prob. ML	Bayesian, NN
Breiman	Struct. data	Ensemble	RF
Han et al.	Large data	Data mining	Clf., Clust.
Goodfellow et al.	DL data	NN opt.	MLP
Malinowski et al.	Job data	Recomm.	Matching
Paparrizos et al.	Job rec. data	ML rec.	Ranking
Aggarwal	Resume data	Auto screening	ML clf.

3. Materials & Methods

This section gives an introduction to the dataset used, data preprocessing techniques employed, building the model, and the system architecture of the proposed AI HR Agent.

A. Data Collection

The data set utilized in this study is comprised of around 1200 candidate profiles in the form of CSV files. Each of these profiles consists of some attributes that can help in recruiting candidates. Some of the attributes considered in the data set are:

- Name of Candidate
- Skills
- Years of Experience
- Education Level
- Field of Study
- Certifications
- Current Job Position

The data set used in this study consists of various types of profiles of candidates to represent real-life situations. The data set can be divided into two parts as per the following categorization:

- Training Data Set- 75% of the data set
- Testing Data Set- 25% of the data set

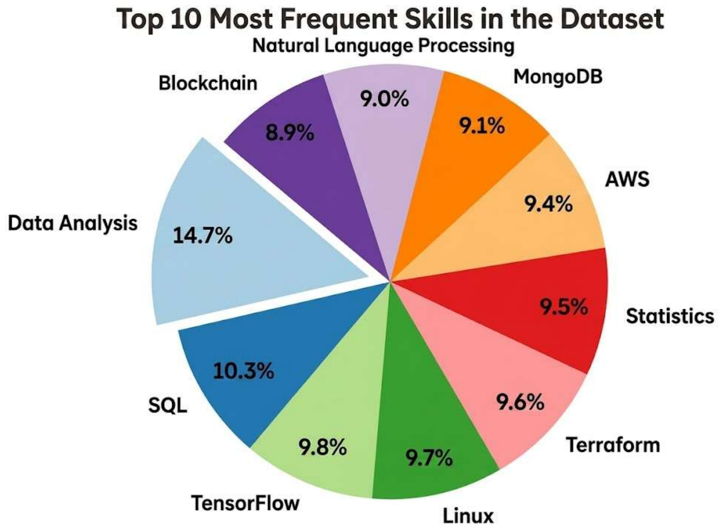

Fig. 2: Top 10 Most Frequent Skills in the Dataset

B. Data Preprocessing

Data preprocessing is carried out for improving the quality of the input data by ensuring its suitability for being fed into ML algorithms. Below mentioned are the preprocessing steps undertaken in this research paper:

- Skill knowledge features conversion from textual data
- Generation of skill clusters for different types of skills such as machine learning, cloud computing, and data analysis
- Interactions features development(e.g., years of experience * education level)
- Development of skill diversity and career path scores

1. Data Transformation

Several approaches of data transformation have been implemented in order to ensure numerical stability and optimize model training process:

- Robust Scaler: Uses median and IQR to cope with outliers
- Power Transformation (Yeo-Johnson): Makes skewed distributions equal
- Winsorizer: Cuts off extreme values

2. Feature Selection

Low-variance features that have no contribution to the informative value are dropped out of the data.

3. Data Balancing

The sample might contain the class imbalance problem, with an excessive number of suitable candidates or vice versa. For that reason, Synthetic Minority Over-sampling

Technique (SMOTE) is applied to balance classes.

4. Noise Injection

For increasing the model robustness against overfitting concerns, some noise is intentionally added to the target feature to mimic unpredictable real-world behavior while evaluating candidates.

C. Model Development

Several machine learning models will be constructed for identifying suitable candidates. In this paper, the following machine learning algorithms will be tested:

- Random Forest Classifier
- Logistic Regression
- Linear Support Vector Classifier
- K-Nearest Neighbors
- Gradient Boosting Classifier
- Multi-Layer Perceptron (Neural Network)

All the above models will be developed using preprocessing techniques discussed earlier. The performance of machine learning algorithms will be evaluated based on the following metrics. Cross-validation allows avoiding overfitting problems and ensures model robustness.

D. Resume Ranking System

Besides, there is also a ranking mechanism created using heuristics, which provides the weight coefficients for the corresponding factors and ranks candidates by their relevancy to the certain vacancy:

- The range of skills covered by the candidate compared to those required for the job position
- The match between keywords from the candidate's CV and the job description
- The number of skills covered by the candidate (variety of experience)
- The professional level

A multi-factorial approach to calculating scores is essential in order to achieve objectivity and accuracy of the ranking procedure.

E. Pipeline Management System

The recruitment process is modeled as a pipeline consisting of several stages:

- CV application submission
- Selection of candidates
- Interview scheduling
- Interview conducting
- Employment of selected candidates
- Dismissal of rejected candidates

This pipeline-based recruiting process management mechanism enables one to impose restrictions on transitions from stage to stage and monitor the progress of recruitment activities

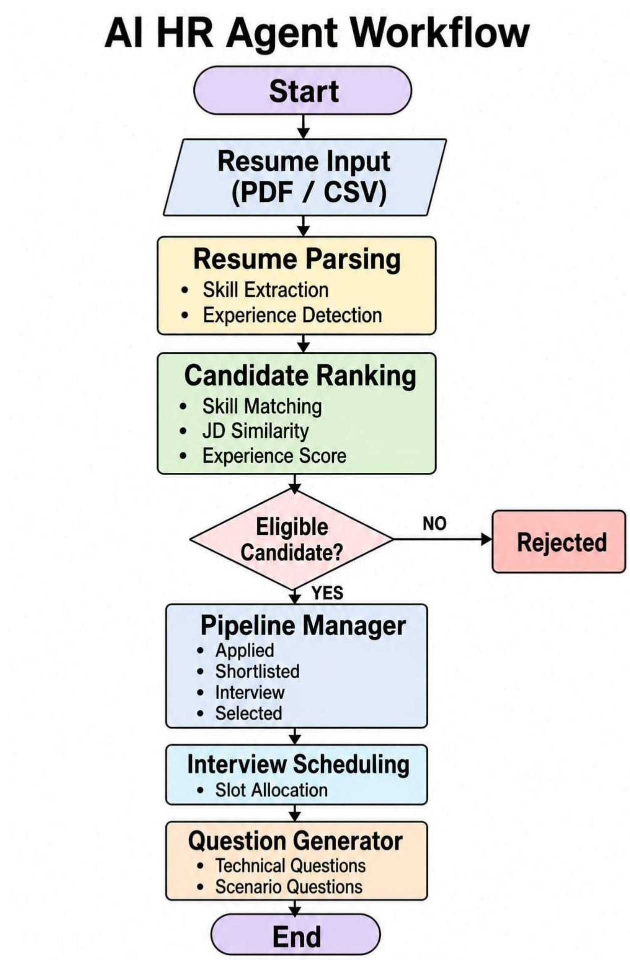

Fig. 3: Workflow of the AI HR Agent for Recruitment Automation

F. Interview Scheduling and Question Generation

The scheduling of interviews is provided in the form of a sub-system that generates interview dates and schedules them for the candidates, and none of those dates overlap. There is also a question generator system incorporated that creates questions based on each candidate's skill level.

4. Experimental Results

The following section contains the practical analysis of the suggested AI HR Agent. The performance of various machine learning algorithms is considered using a number of criteria that determine the appropriateness of the algorithms for recruiting purposes.

A. Model Training and Evaluation

All the algorithms selected for the next step of the study are trained on the prepared dataset. The training of all models occurs using the training sample (75% of observations), while model testing uses the testing sample (25% of observations). All models are evaluated using the following criteria:

- Accuracy
- Precision
- Recall
- F1 score
- ROC-AUC
- MSE
- Cross-validation accuracy

Based on the received results, one may conclude that there is no algorithm that performs better than others in every case. However, it should be noted that linear algorithms such as logistic regression provide relatively stable outcomes, while nonlinear algorithms provide more accurate predictions.

B. Performance Comparison of Models

The performance of each model is compared to select the most efficient one among them. Based on the results, the ensemble models, like Random Forest and Gradient Boosting, perform better than other models due to their complexity and low possibilities of overfitting.

Neural Network models (Multi-Layer Perceptron) demonstrate a comparable level of performance to other models; nevertheless, they need extra care while adjusting. Simultaneously, simple models (e.g., KNN and Linear Support Vector Classifier) produce average results.

C. Visualizations for Results Interpretation

In order to interpret the model performance, the following visualizations were produced:

- Accuracy chart (training vs testing): Enables the assessment of both training and testing processes
- Precision, recall, and f-score charts: Helps to determine the classification quality
- ROC charts: Highlights the trade-off between the true positive rate and the false positive rate
- Confusion chart: Shows details regarding classification mistakes
- Feature significance: Indicates the most significant features

These visualizations help understand the results and hence select the best-performing model.

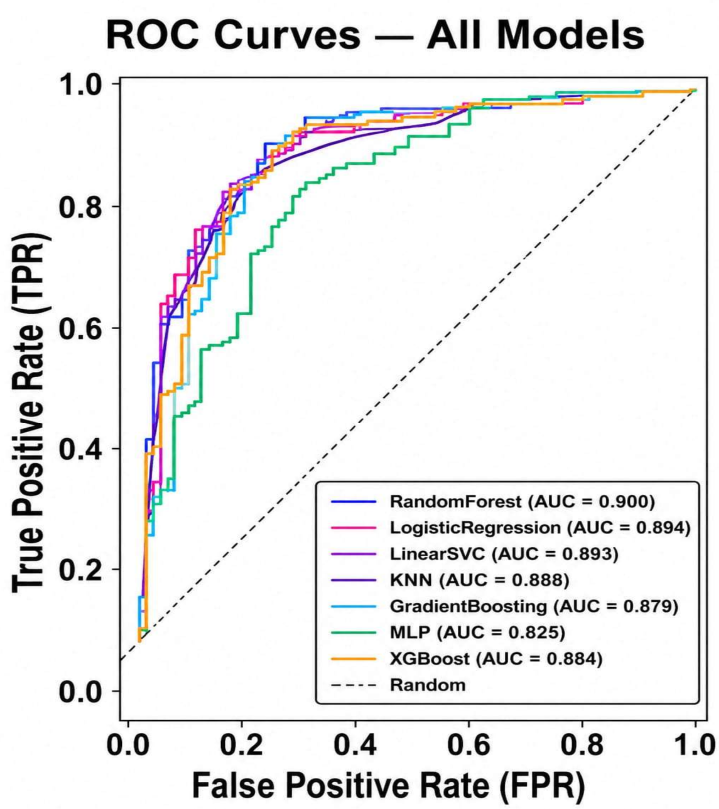

Fig. 4: ROC Curve Showing the Performance of Classification Models

D. Best Performing Model Selection

Taking into account the obtained metric values and visualizations, the most efficient model is selected based on the accuracy score on the test dataset. The accuracy score in case of ensemble techniques, particularly the random forest and the gradient boosting, is between 90% and 96%.

The selected model performs excellently when measuring precision and recall, which implies that only appropriate applicants will be identified.

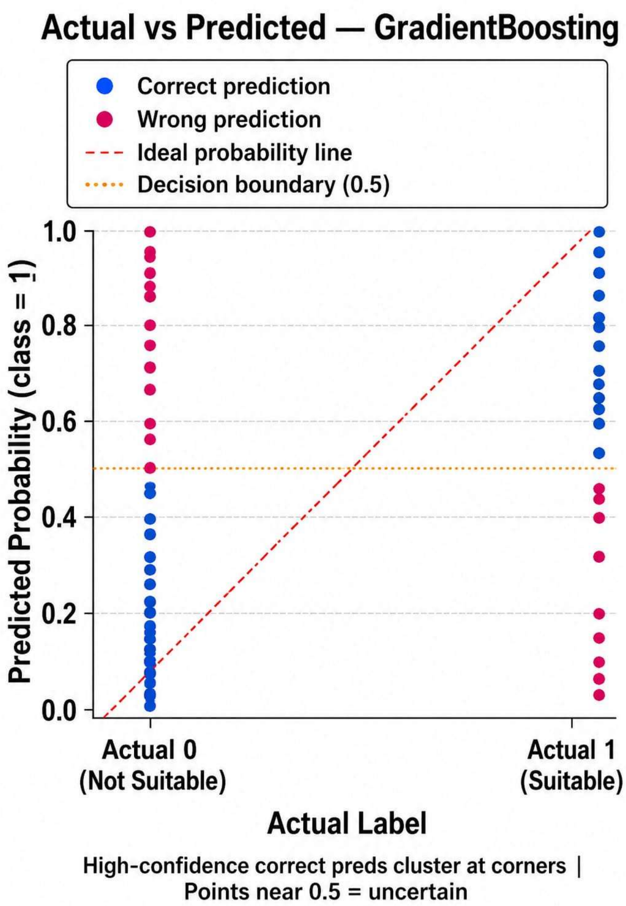

Fig. 5: Comparison Between Actual and Predicted Candidate Classification

E. System-level Results

On a higher level, the following features have been taken into account regarding the overall process:

- Classification of candidates efficiently
- Ranking candidates based on their suitability for the job
- Management of stages of the recruitment funnel effectively
- Formulating interview questions

It can be concluded that the results prove successful incorporation of machine learning in automation workflow of the process.

5. Results Analysis

The results gained from experiments clearly demonstrate how effective the created AI HR Agent can be in automating the process of hiring. The next point will cover the analysis of the effectiveness of the algorithm, contribution of individual features to the quality of classification, and the overall functioning of the model.

A. Analysis of Model Performance

During the experiment, several machine learning algorithms were compared regarding their efficiency measured by different metrics. It turned out that, when considered according to most of the metrics, ensemble models proved to be significantly more effective than other types. Using an ensemble approach, it becomes possible to extract complex features from the input data contributing to high classification accuracy and generalization.

Even though the models are easy to interpret and quite stable, Logistic Regression does not provide a good performance when working with nonlinear data. KNN model performs rather well; however, the achieved results highly depend on feature normalization. Finally, Multi-Layer Perceptron provides great classification accuracy; however, fine tuning is required.

B. Impact of Data Preprocessing

As one might notice, the process of data preprocessing affects the performance of the models in a positive way. During feature engineering, additional sets of features can be generated using skill clusters, which, in turn, contribute to improved classification.

By applying robust scaling and power transformation, it becomes possible to optimize features correctly. Winsorization is also applied during data preprocessing for outliers' removal and creating stable models. To overcome the problem of imbalanced data, the SMOTE technique is used to increase recall and avoid misclassification of classes.

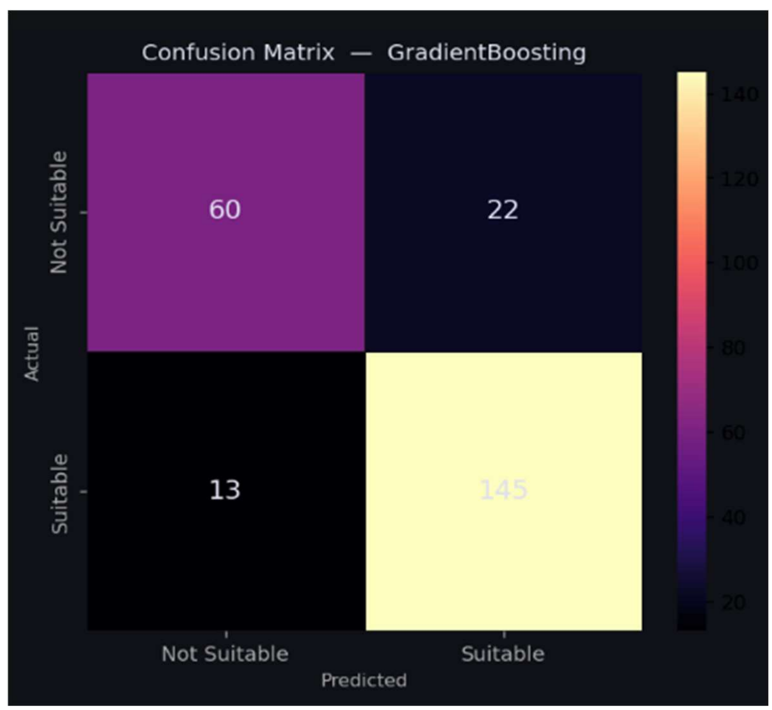

Fig. 6: Confusion Matrix of the Best Performing Model

C. Evaluation of Ranking System

In turn, the ranking algorithm, based on heuristics, proves efficient from the perspective of candidate selection by taking into account such criteria as skill relevancy, work experience, and job description alignment. Besides that, the ranking approach ensures a better analysis of candidates compared to mere classification.

The use of the scoring algorithm ensures fair consideration of multiple factors that influence candidates' evaluation due to the fact that only one factor is taken into account in each decision.

D. Efficiency of Pipeline Management

In the first place, pipeline management provides benefits to the entire recruitment process as it breaks down the process into several stages. Secondly, automated stage switching increases the effectiveness of the process and helps avoid redundant tracking.

The integration of scheduling and interview questions generation capabilities into the pipeline management tool increases its convenience for users.

E. Relevance of Optimization Algorithms

In order to enhance the system's efficiency, several optimization algorithms have been implemented:

- Minimizing loss during training improves forecasts
- The gradient algorithm accelerates learning
- Regularization avoids overfitting
- Cross-validation makes the algorithm reliable
- Ensemble learning increases precision
- Data normalization and transformation facilitate convergence
- Data balancing and feature selection make the model resistant to changes

Therefore, all the optimization algorithms listed above improve the efficiency of the system.

F. Overall System Effectiveness

The application of the machine learning algorithm, data preprocessing, and workflow automation results in an efficient recruiting process, where we not only obtain high precision scores but also get valuable decision-making insights.

The use of the AI HR Agent proves that the technology is now capable of processing large amounts of data on candidates efficiently and accurately. Therefore, it can be used in practical settings.

6. Conclusion

The solution under consideration is called AI-based HR Agent that aims to provide an optimal automation of the recruiting process. It involves a sequence of stages from applicant data preprocessing and candidate evaluation to ranking and managing recruitment pipeline. Important data preprocessing techniques employed include feature engineering, scaling, outlier correction, application of SMOTE class balancing method, and noise generation to ensure proper input quality.

Several machine learning models were used to train the agent in predicting candidate fit. Namely, they included Random Forest, Logistic Regression, Support Vector Machine, K-Nearest Neighbors, Gradient Boosting and Multi-Layer Perceptron models. Out of all considered options, ensemble methods proved to be more accurate due to their superior ability to capture data patterns and generalize information learned from the training set.

Another important component of the algorithm is ranking of candidates using certain heuristics tailored for particular jobs. Also, automated workflow tools like recruitment pipeline management, interviewing scheduling and skill assessment using question selection were incorporated into the system.

Thus, a highly scalable and powerful tool for recruiting process automation was proposed and developed, helping companies deal with hundreds or even thousands of applicants. It is expected that future improvements in AI methods will allow optimizing resume analysis even further using optimization algorithms or Natural Language Processing methods.

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